

WELL COMPLETION REPORT (SYNTHETIC)

OIL-CHR-004 - Charali-004

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000047
Field / District	Charali / Biswanath
Block	PEL-ASSAM-08
Basin	Assam-Arakan
Formation	Girujan (GRJ)
Spud / Completion	2005-12-06 / 2006-03-09
Measured / TVD	2108 m / 2054 m
Trajectory	Directional; inc 34.2 deg; az 325.2 deg
Formation Interval	997-1545 m
Lithology	Silty shale with sandstone lenses
Pore Pressure Reference	1943 psi
Fracture Gradient	14.64 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	Low (39.6/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	527	Surface
9 5/8 in casing	1222	Intermediate
7 in liner/casing	1855	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2005-12-30 670 m	Kick	Critical	Unexpected background gas rise preceded influx. Mitigation: Increase mud weight within fracture-margin limits and monitor gas trend. Lesson: Treat pit gain plus gas increase as an early influx indicator.
2006-01-21 1145 m	Stuck Pipe	High	Torque and drag increased while ROP declined. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2006-02-01 1363 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2006-02-15 1652 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2006-01-05 791 m	Stuck Pipe	Medium	Hookload became erratic during pipe movement. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2006-01-11 924 m	Torque Spike	Medium	Torque rose without corresponding pressure anomaly. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Use offset torque trend to set adaptive operating envelope.
2006-01-19 1094 m	NPT	Medium	No depth progress occurred during the recorded interval. Mitigation: Maintain critical spares for recurring equipment failures. Lesson: Separate geological risk from equipment-driven NPT in analytics.

Completion Summary

The synthetic well reached 2108 m MD in the Girujan interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.